

**Department of Computer Technology****Vision of the Department**

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, - National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027(ODD)

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)



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Session	2026-27 (ODD)	Course Name	LAB: Software Engineering
Semester	7th	Course Code	23CT1704
Roll No	83	Name of Student	Trupal Tagade

Practical Number	1
Aim	Introduction to Software Engineering fundamentals, UML and Star UML
Theory (100 words)	1. Software Engineering Software Engineering is a systematic approach to develop, test, deploy and maintain software. Main activities: ● Requirement Analysis ● Design ● Coding ● Testing ● Deployment ● Maintenance 2. UML UML = Unified Modeling Language UML is a standard visual language used to model and design software systems. Uses: ● Understand system structure



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- Show system behavior
- Improve communication
- Document software design

3. UML Diagram Types

Structural:

- Class Diagram
- Object Diagram
- Component Diagram
- Deployment Diagram

Behavioral:

- Use Case Diagram
- Sequence Diagram
- Activity Diagram
- State Diagram

4. StarUML

StarUML is a software tool used to create UML diagrams.

Features:

- Easy to create UML diagrams
- Supports Class, Use Case, Sequence, Activity diagrams
- Helps in software analysis and design
- Provides graphical modeling of systems



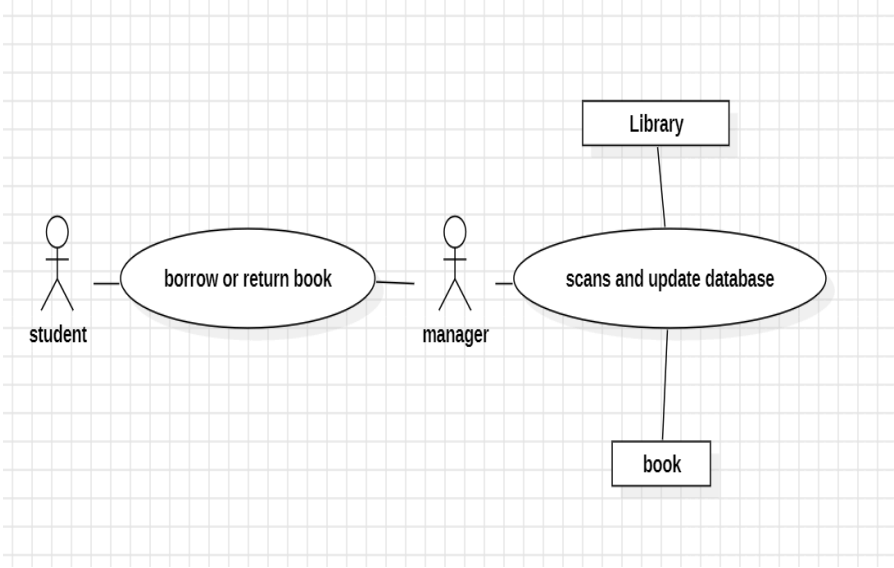
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Procedure and Execution (100 Words)	Algorithm(Steps): 1. Create a new workspace. 2. Create a Use Case Diagram. 3. Add actors: Student and Manager. 4. Add use cases: Borrow/Return Book and Scan & Update Database. 5. Add entities/classes: Library and Book. 6. Connect the actors and use cases using appropriate relationships.
Output Analysis	
Link of student Github profile where lab assignment has been uploaded	23070254-sys
Conclusion	The Library Management System diagram represents the basic interaction between the student, manager, library, and



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	book. It shows how students borrow or return books and how the manager scans books and updates the database.
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